

SP ENGINEERING ACADEMY

Video Lectures / Course Materials

COMPLETE COURSE

DIGITAL SYSTEMS

DIGITAL CIRCUIT DESIGN

Part – III - Binary Arithmetic

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Binary Arithmetic

1. Binary Addition

Video Link - <https://youtu.be/476LmYDxtJw>

The addition consists of four possible elementary operations as shown below

$$1 + 0 = 1$$

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 1 = 10$$

Inputs			Outputs	
A	B	Cin	Sum	Out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Example 1

Perform addition of $(11001100)_2$ and $(11011010)_2$

Augend Carry $\rightarrow$ 1 1 1 0 0 1 1 0 0 $\leftarrow$ Number 1
Addend Numbers $\rightarrow$ 1 1 0 1 1 0 1 0 $\leftarrow$ Number 2
Sum Number 2 $\rightarrow$ 1 1 0 1 0 0 1 1 0 $\leftarrow$ Result
Result $\rightarrow$

110100110 - Answer

Example 2

Add $(28)_{10}$ and $(15)_{10}$ by converting them into binary

$$(28)_{10} \rightarrow 11100$$

$$(15)_{10} \rightarrow (1111)_2$$

$$\begin{array}{r} 2 \overline{) 28} \\ 2 \overline{) 14} - 0 \\ 2 \overline{) 7} - 0 \\ 2 \overline{) 3} - 1 \\ 1 - 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 15} \\ 2 \overline{) 7} - 1 \\ 2 \overline{) 3} - 1 \\ 1 - 1 \end{array}$$

$$\text{Carry} \rightarrow 111$$

$$\text{Binary equivalent of } 28 \rightarrow 011100$$

$$\text{Binary equivalent of } 15 \rightarrow 001111$$

$$\text{Result} \rightarrow 101011$$

$$\text{Answer } (101011)_2$$

2. Binary Subtraction [Video Link - https://youtu.be/cVmesUaGrTE](https://youtu.be/cVmesUaGrTE)

$$0 - 0 = 0$$

$$0 - 1 = 1 \text{ with } 1 \text{ borrow}$$

$$1 - 0 = 1$$

$$1 - 1 = 0$$

Example 1

$$\text{Perform } (11101100)_2 - (00110010)_2$$

$$\begin{array}{r} 0 10 10 10 \\ \text{Number 1} \rightarrow 1 \cancel{1} 0 1 \cancel{0} 0 \text{ Minuend} \\ \text{Number 2} \rightarrow 00110010 \text{ Subtrahend} \\ \text{Result} \rightarrow 10111010 \text{ Difference} \end{array}$$

$$\text{Answer } (10111010)_2$$

$$(10)_2 - (1)_2 = (1)_2$$

The Subtraction technique we have just seen is difficult to implement in a computer. Thus, computer uses 2's Complement.

(a) Binary Subtraction Using 1's Complement method

Video Link - <https://youtu.be/UcefVVhVr8>

In a 1's Complement Subtraction, negative number is represented in the 1's Complement form and actual addition is performed to get the desired result.

$A - B$ is performed using following steps

→ Take 1's Complement of B

→ Result $\leftarrow A + 1's \text{ Complement of } B$

→ If Carry is generated then the result is positive and in the true form. Add Carry to the result to get the final result.

→ If Carry is not generated then the result is negative and in the 1's Complement form.

Example 1

Perform $(28)_{10} - (15)_{10}$ using 6-bit 1's Complement representation.

$$(28)_{10} \rightarrow (011100)_2 \quad (15)_{10} \rightarrow (001111)_2$$

$$\begin{array}{r} 2 \overline{) 28} \\ 2 \overline{) 14} - 0 \\ 2 \overline{) 7} - 0 \\ 2 \overline{) 3} - 1 \\ \underline{1 - 1} \end{array}$$

$$\begin{array}{r} 2 \overline{) 15} \\ 2 \overline{) 7} - 1 \\ 2 \overline{) 3} - 1 \\ \underline{1 - 1} \end{array}$$

$$(15)_{10} \rightarrow 001111$$

1's Complement of $(15)_{10} \rightarrow 110000$

$$\begin{array}{r}
 \text{Carry} \rightarrow 1 \ 1 \\
 \text{Binary eqn of } (28)_{10} \rightarrow 0 \ 1 \ 1 \ 1 \ 0 \ 0 \\
 \text{1's Complement of } (15)_{10} \rightarrow 1 \ 1 \ 0 \ 0 \ 0 \ 0 + \\
 \quad \quad \quad 1 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0 \\
 \quad \quad \quad \underline{\hspace{1cm} \rightarrow 1 \hspace{1cm}} \\
 \text{Final Result} \quad \quad \quad 0 \ 0 \ 1 \ 1 \ 0 \ 1
 \end{array}$$

$$\underline{(001101)_2} \rightarrow (13)_{10}$$

$$\begin{array}{cccccc}
 0 & 0 & 1 & 1 & 0 & 1 \\
 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0
 \end{array}$$

$$8 + 2 + 1 = 13$$

$$\begin{array}{r}
 + 28 \\
 - 15 \\
 \hline
 13
 \end{array}$$

Example 2

Perform $(15)_{10} - (28)_{10}$ Using 6-bit -
 1's Complement representation

$$(15)_{10} \rightarrow (001111)_2$$

$$(28)_{10} \rightarrow (011100)_2$$

$$\text{1's Complement of } (28)_{10} \rightarrow (100011)_2 \leftarrow$$

$$\begin{array}{r}
 \text{Carry} \rightarrow \quad \quad \quad 1 \ 1 \ 1 \ 1 \\
 \text{Binary eqn of } (15)_{10} \rightarrow 0 \ 0 \ 1 \ 1 \ 1 \ 1 \\
 \text{1's Complement of } (28)_{10} \rightarrow 1 \ 0 \ 0 \ 0 \ 1 \ 1 + \\
 \text{Result} \rightarrow \quad \quad \quad 1 \ 1 \ 0 \ 0 \ 1 \ 0
 \end{array}$$

$$\text{1's Complement of Result} \rightarrow 001101$$

$$\text{Final Answer} \quad \underline{001101 = (13)_{10}}$$

(b) Binary Subtraction using 2's Complement Method

Video Link - <https://youtu.be/t0L7mkl77w>

In a 2's Complement Subtraction, negative number is represented in the 2's Complement form and actual addition is performed to get the desired result.

$A - B$ is performed using following steps

→ Take 2's Complement of B

→ Result $\leftarrow A + 2$'s Complement of B

→ If Carry is generated then the result is positive and is in the true form. In this case, Carry is ignored

→ If Carry is not generated then the result is negative and in the 2's Complement form

Example 1

Perform $(28)_{10} - (15)_{10}$ Using 6-bit 2's Complement representation

$$(28)_{10} \rightarrow 001111$$

$$(28)_{10} \rightarrow 011100$$

$$(15)_{10} \rightarrow 001111$$

$$1\text{'s Complement of } (15)_{10} \rightarrow 110000$$

$$2\text{'s Complement of } (15)_{10} \rightarrow \text{Add } 1 \rightarrow 110000$$

$$110001$$

$$\begin{array}{rcl}
 \text{Carry} & & 1 \\
 \text{Binary eqn } (28)_{10} & \rightarrow & 011100 \\
 2\text{'s Compl of } (15)_{10} & \rightarrow & 110001 \\
 \text{Result} & \rightarrow & 1001101
 \end{array}$$

Ignore Carry
Final Answer $001101 \rightarrow (13)_{10}$

Example 2

Perform $(15)_{10} - (28)_{10}$ using 2's Complement representation

$$\begin{array}{l}
 (15)_{10} \rightarrow (001111)_2 \\
 (28)_{10} \rightarrow (011100)_2
 \end{array}$$

$$1\text{'s Complement of } (28)_{10} \rightarrow 100011$$

$$2\text{'s Complement of } (28)_{10} \rightarrow \text{Add } 1 \rightarrow 100100$$

$$\begin{array}{rcl}
 \text{Carry} & & 111 \\
 \text{Binary eqn of } (15)_{10} & \rightarrow & 001111 \\
 2\text{'s Compl of } (28)_{10} & \rightarrow & \text{000100} + \\
 \text{Result} & \rightarrow & 10011
 \end{array}$$

No Carry, thus result is negative and in 2's Complement form

$$\begin{array}{rcl}
 1\text{'s Complement of result} & & 001100 \\
 2\text{'s Complement} \rightarrow \text{Add } 1 & & 1 \\
 & & 001101
 \end{array}$$

$$\text{Final Result} - (001101)_2 \rightarrow (13)_{10}$$

Example 3

Video Link - <https://youtu.be/3R2qRWK8Wdg>

Perform $01000 - 01001$ using 2's Complement method

1's Complement is 10110
2's Complement $\rightarrow$ Add 1

$$\begin{array}{r} 01000 \\ 10111 + \\ \hline 11111 \end{array}$$

No Carry, result is negative and in 2's Complement form

$\therefore$ 1's Complement of result 00000
2's Complement $\rightarrow$ Add 1

00001

Final Answer $(00001)_2$

Example 4

Perform $0011.1001 - 0001.1110$ Using 2's Complement form

Subtractor 0001.1110
1's Complement of Subtractor 1110.0001
2's Complement $\rightarrow$ Add 1

1110.0010

Carry $\rightarrow 11$

$$\begin{array}{r} 0011.1001 \\ 1110.0010 + \\ \hline 10001.1011 \end{array}$$

Ignore Carry

Final answer $(0001.1011)_2$

Example 5

Video Link - <https://youtu.be/EX3dG2uVQLg>

Perform $(-42) - (-13)$ using 2's Complement representation.

$$(-42) - (-13) = -42 + 13$$

$$(13)_{10} \rightarrow (0001101)_2$$

$$(42)_{10} \rightarrow (0101010)_2$$

$$\begin{array}{r} 2 \overline{) 13} \quad 2 \overline{) 42} \\ 2 \overline{) 6} \quad 2 \overline{) 21} - 0 \\ 2 \overline{) 3} - 0 \quad 2 \overline{) 10} - 1 \\ 1 - 1 \quad 2 \overline{) 5} - 0 \\ \quad 2 \overline{) 2} - 1 \\ \quad \quad 1 - 0 \end{array}$$

$$1's \text{ Complement of } (42)_{10} \rightarrow 1010101$$

$$2's \text{ Complement} \rightarrow \text{Add } 1$$

$$1010110$$

$$\text{Carry} \rightarrow 111$$

$$(13)_{10} - \text{Binary eq} \rightarrow 0001101$$

$$2's \text{ Compl of } (42)_{10} \rightarrow 1010110 + 1100011$$

No Carry, result is negative and in 2's Complement form

$$1's \text{ Complement} \rightarrow 0011100$$

$$2's \text{ Complement} \rightarrow \text{Add } 1$$

$$\underline{0011101}$$

Multiplicand

Multiplier

Product

Dividend

Divisor

Quotient / Remainder

Complement of Numbers

Video Link - <https://youtu.be/htmuoToA0T4>

Complement numbers are used in case of Subtraction of two numbers in corresponding number Systems.

(a) 1's Complement Representation

The 1's Complement of a binary number is the number that results when we change all 1's to Zeros and Zeros to ones

~~(a)~~ 1's Complement of 11010100 is 00101011

(b) 2's Complement Representation

The 2's Complement is the binary number that results when we add 1 to the 1's Complement.

2's Complement of $(11000100)_2$ is

1's Complement is 00111011

2's Compl $\rightarrow$ Add 1

$(00111100)_2$

Example

Represent the decimal number -200 and 200 Using 2's Complement binary form

To represent 200 and -200 we need 9-bit number system

$+200 = 011001000$

$-200 = 100110111$ - 1's Complement
Add 1

100111000 - 2's Complement

Signed Binary Numbers

Representation of Signed numbers using 2's or 1's Complement method

If the number is positive, the magnitude is represented in its binary form and a sign bit 0 is placed in front of MSB

Example

Represent +51 and -51 in 2's Complement

$$+51 \Rightarrow 00110011$$

$$-51 \Rightarrow 10110011$$

$$1's \text{ Complement of } -51 \Rightarrow 11001100$$

$$2's \text{ Complement of } -51 \Rightarrow 11001101$$

$$\begin{array}{r} 2 \overline{) 51} \\ 2 \overline{) 25-1} \\ 2 \overline{) 12-1} \\ 2 \overline{) 6-0} \\ 2 \overline{) 3-0} \\ 1-1 \end{array}$$

Decimal Subtraction

① Perform 841 - 329

Video Link - https://youtu.be/T_9HsHOzUa8

Normal Subtraction

$$\begin{array}{r} 841 \\ 329 (-) \\ \hline 512 \end{array}$$

9's Complement

$$\begin{array}{r} 9's \text{ Complement of } 329 \text{ is } 999 \\ 329 \\ \hline 670 \end{array}$$

Number 1 $\rightarrow 841$

9's Comp of 329 $\rightarrow 670$

$$\begin{array}{r} 1511 \\ \rightarrow 1 \\ \hline 512 \end{array}$$

Carry generated
Answer is positive
Add carry to get
final answer

10's Complement

$$\begin{array}{r} 999 \\ 329 \\ \hline 670 \end{array}$$

$$\begin{array}{r} 10's \text{ Complement is } 670 \\ 1 \\ \hline 671 \end{array}$$

$$\text{Number 1} \rightarrow 841$$

$$\begin{array}{r} 10's \text{ Compl of } 329 \rightarrow 671 \\ \hline 1512 \end{array}$$

Carry is generated so the result is positive and in true form

② Perform $329 - 841$

Video Link - <https://youtu.be/3yu95zlpIMY>

Normal Subtraction

$$\begin{array}{r} 841 \\ 329 \\ \hline - 512 \\ \hline \end{array}$$

9's Complement

$$9's \text{ Complement of } 841 \rightarrow 999$$

$$\begin{array}{r} 841 \\ \hline 158 \\ \hline \end{array}$$

$$\text{Number 1} \rightarrow 329$$

$$\begin{array}{r} 9's \text{ comp of } 841 \rightarrow 158 \\ \hline 487 \end{array}$$

No Carry, So answer is negative and in 9's Complement form

$$\begin{array}{r} 999 \\ 487 \\ \hline - 512 \rightarrow \text{Final Answer} \end{array}$$

10's Complement

$$\begin{array}{r} 999 \\ 10's \text{ Complement of } 841 \rightarrow \underline{841} \\ 158 \\ 10's \text{ Complement of } 841 \rightarrow \underline{158} \\ 1 \\ \hline 159 \end{array}$$

Number 1 $\rightarrow 329$

$$10's \text{ Compl of } 841 \rightarrow \underline{159} \\ 488$$

No Carry so answer is negative
and 10's Complement form

$$\therefore 9's \text{ Complement of } 488 \rightarrow \begin{array}{r} 999 \\ \underline{488} \\ 511 \end{array}$$

$$10's \text{ complement of } 488 \rightarrow \begin{array}{r} 511 \\ 1 \\ \hline 512 \end{array}$$

$\therefore$ Final answer = 512

③ Perform $8486 - 765$

Video Link - <https://youtu.be/nLHGBi6hdK4>

Normal Subtraction

$$\begin{array}{r} 8486 \\ - 765 (-) \\ \hline 7721 \end{array}$$

9's Complement

$$\begin{array}{r} 9999 \\ - 765 (-) \\ \hline 9234 \end{array}$$

$$\begin{array}{r} \text{Number 1} \rightarrow 8486 \\ 9's \text{ Compl of } 765 \rightarrow 9234 (+) \\ \hline 17720 \\ \text{L} \rightarrow 1 (+) \\ \hline 7721 \end{array}$$

Carry is generated, answer is positive and add the carry to get final answer

10's Complement

$$\begin{array}{r} 9999 \\ - 765 (-) \\ \hline 9234 \\ \text{10's Complement is} \quad \quad \quad \hline 1 (+) \\ \hline 9235 \end{array}$$

$$\begin{array}{r} \text{Number 1} \rightarrow 8486 \\ 10's \text{ Compl of } 765 \rightarrow 9235 (+) \\ \hline 17721 \\ \hline \end{array}$$

Carry is generated, So Answer is positive and in true form. Ignore the Carry.